Task 2 - Choose the Data Types

C Programming Assignment - Rural Electricity Billing System

Recommended types

Use int64_t for money in paise and signed intermediate totals, uint64_t for non-negative cumulative meter readings, enum for categories and size_t for array sizes. Include stdint.h and inttypes.h.

Why integer paise

Store ■123.45 as 12345 paise. This avoids binary floating-point rounding in currency. Display rupees using amount divided by 100 and paise using amount modulo 100.

Type limits

C does not guarantee identical sizes for int, long and long long on every platform. Use sizeof and limits from limits.h or stdint.h. Signed overflow is undefined behavior; check additions and multiplications.

16-bit rollover

A 16-bit unsigned meter ranges from 0 to 65535. If rollover is confirmed, consumed = (65535 - previous) + 1 + current. Previous 65530 and current 5 gives 11 counts. Never assume rollover without checking the meter specification.

Precision

0.1 + 0.2 may not be represented exactly as a binary float. Integer arithmetic is exact until overflow.

Important: illustrative tariff rates are used because the screenshot does not show the complete official tariff table. Replace them with your instructor's prescribed rates.